Choose a motion method

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Learning target: Choose a motion relationship and justify its assumptions.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 1, Day 5; 3 course days later.*
Central position 13 m; interval 11-15 m. Assumes steady velocity and the stated start.

02. Explain a worked example (worked)
Triangle base 3 s, height 6 m/s; area half of 18=9 m.

03. Choose the relationship (model)
Steady velocity; 3×4=12 m.

04. Build final velocity (practice)
Gain 6 m/s; final 8 m/s east.

05. Find the distance while speeding up (reason)
Average 5 m/s east; displacement 15 m east.

06. Choose a method (mixed) *Review: Unit 1, Day 7; 1 course days later.*
Slowing down; change is 7 m/s east. Acceleration 3.5 m/s² east.

07. Exit check (exit)
a=2 m/s² east; average 3 m/s east; displacement 9 m east.

08. STRETCH: Apply it another way (stretch)
Steady 2 m/s; accelerating from 0 to 4 m/s. Both graph areas 8 m.

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01. Bring back an earlier idea (retrieve) *Review: Unit 1, Day 5; 3 course days later.*
Central 32 m; half-width 1.2 m; interval [30.8, 33.2] m; full width 2.4 m. Constant velocity and bounded input ranges.

02. Explain a worked example (worked)
Triangle base 3 s, height 6 m/s; area half of 18=9 m.

03. Solve and check (model)
v=6 m/s;Δx=8+8=16 m; final x=13 m.

04. No time given (practice)
0=144-6Δx;Δx=24 m; time 4 s; average 6 m/s×4=24 m.

05. At the top (reason)
Rise 10 m; time 14/9.8=1.43 s; acceleration remains 9.8 m/s² down.

06. Choose a method (mixed) *Review: Unit 1, Day 7; 1 course days later.*
First a=-4 m/s², speeding up. Then a=0, constant speed 15 m/s.

07. Exit check (exit)
Stops at 4 s after 24 m. Continued negative acceleration after rest would reverse it; ordinary braking ends the model at the stop.

08. STRETCH: Apply it another way (stretch)
24=8+8 a gives a=2 m/s². Average v=6, final v=10, so a=(10-2)/4=2. Initial speed alone is not average speed.